EXECUTIVE BUSINESS WHITEPAPER

GCMS Clinical AI Consultation Platform

Secure, Local Decision-Support Architecture for Singapore Primary Care

System Name:	GCMS Clinical AI Consultation Platform
Version / Edition:	Version 1.2.0 (Singapore Healthcare Edition)
Target Audience:	Chief Medical Officers, Medical Directors, Clinic Owners, Regulators
Regulatory Scope:	Singapore MOH Cybersecurity Guidelines, HSA AI-MD, Singapore PDPA
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1. Executive Summary

The GCMS Clinical AI Consultation Platform is an enterprise-grade clinical decision support solution designed specifically for outpatient clinics and polyclinics in Singapore. The system addresses critical healthcare challenges: physician documentation burden, diagnostic complexity in high-volume clinics, and the severe medicolegal and regulatory risks associated with sending patient data to public cloud AI APIs.

By utilizing local open-weights language models through Ollama, the system executes 100% of reasoning within the local clinic infrastructure or private VPC. No patient data leaves the approved healthcare boundary. The platform couples this local AI reasoning with licensed SNOMED CT and Singapore Drug Dictionary (SDD) terminologies to guarantee zero-hallucination medical coding.

> **[SECURITY] Doctor-in-the-Loop Clinical Principle**
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> The system operates strictly as decision support. The AI never finalizes orders, writes prescriptions autonomously, or substitutes clinician judgment. Every AI suggestion requires review and doctor confirmation before recording into the Clinic Management System.

2. Strategic Value Proposition & ROI

Operational Metric	Traditional Consultation	GCMS AI-Assisted Consultation
Encounter Documentation	Physicians spend 4-7 minutes per patient typing free-text notes.	**Reduced by 35%** with automated clinical extraction and structured suggestions.
Terminology Coding	Manual search across 350,000+ SNOMED CT concepts.	**Instant sub-second semantic search** and automated diagnostic code assignment.
Prescription Safety	Manual recall of drug interactions and patient allergies.	**Automated deterministic safety engine** checking cross-allergies and contraindications.
Data Sovereignty	Commercial cloud AI APIs send PHI over public networks.	**100% On-Premises data residency**; zero outbound transmission.
Operating Cost	Recurring monthly per-token API charges ($0.03-$0.10/call).	**Zero recurring token fees**; runs on fixed local hardware.

3. Core Business Services

- **Service 1: Licensed SNOMED CT Terminology Service** — High-speed semantic search across Singapore & International editions, standardizing disease classification for MOH NEHR submissions and insurance claims.
- **Service 2: Singapore Drug Dictionary (SDD) Service** — Instantly resolves HSA-registered pharmaceutical brands and active ingredients, ensuring accurate electronic ordering and pharmacy dispensing.
- **Service 3: AI Differential Diagnosis Support** — Synthesizes free-text consultation notes, vital signs, and patient history into ranked differential diagnoses with supporting evidence and missing examination alerts.
- **Service 4: AI Medication Candidate Recommendation** — Suggests guideline-aligned therapeutics for doctor-confirmed diagnoses, automatically matching items against the authorized Singapore Drug Dictionary.

4. Clinical Safety & Regulatory Compliance

Safety is an architectural guarantee rather than an afterthought. The platform decouples generative reasoning from safety-critical rules:

> **[IMPORTANT] Independent Deterministic Safety Engine**
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> Drug-drug interactions, allergy cross-reactivity (e.g., Penicillin to Augmentin), and contraindications (pregnancy, renal impairment) are evaluated by pure, deterministic rule engines in code — NEVER by probabilistic generative AI models.

- **Singapore PDPA Compliance** — A dedicated Privacy Boundary automatically detects and redacts Singapore NRIC/FIN numbers, contact numbers, emails, and patient names before clinical narrative reaches the AI model.
- **Auditability & Governance** — Every inference request, prompt version, model identifier, and clinician interaction is recorded in an immutable audit log, providing complete medicolegal traceability.
- **HSA AI-MD Alignment** — Designed in conformance with Health Sciences Authority guidelines for Artificial Intelligence-based Medical Devices, emphasizing clinician oversight, transparent reasoning, and missing-data flags.